

Stacks & Queues: 50 High-Yield Questions

Comprehensive Revision Bank for GATE, ISRO, BARC, DRDO, and NIC Technical Exams

This handpicked collection focuses on core theoretical mechanics, data structure properties, programmatic conversion rules, and recent 5-year competitive exam question patterns.

1. Stack Conceptual & Properties

Q1. What is the fundamental operational principle of a Stack data structure?

Answer: LIFO (Last In First Out). The element added last is the first one to be removed.

Q2. What error occurs when attempting to remove an element from an empty stack?

Answer: Stack Underflow.

Q3. What condition indicates a "Stack Full" state if two stacks are implemented using a single array $A[1..MAXSIZE]$ growing from opposite ends?

Answer: $top1 + 1 == top2$ (assuming $top1$ starts at 0 and $top2$ starts at $MAXSIZE + 1$).

Q4. Which data structure is explicitly used by the operating system to manage standard recursive function calls?

Answer: System Stack (or Call Stack).

Q5. If the push and pop operations take $O(1)$ time, what is the time complexity to find the minimum element in a standard unmodified stack?

Answer: $O(n)$, as we have to scan all elements by popping them out sequentially.

Q6. How can you augment a stack to support a $getMin()$ operation in $O(1)$ time complexity?

Answer: By maintaining an auxiliary tracking stack that stores the minimum element corresponding to each state of the primary stack.

Q7. Consider the sequence of elements 1, 2, 3, 4. If they are pushed into a stack in this order, which of the following is an invalid permutation output sequence? (A) 1, 2, 3, 4 (B) 4, 3, 2, 1 (C) 3, 1, 2, 4 (D) 2, 3, 4, 1

Answer: (C). You cannot pop 1 before 2 if 3 has already been popped. *Shortcut rule:* An output permutation can never contain a larger element followed by two smaller elements in ascending order.

Q8. What is the minimum number of stacks required to systematically implement a standard Queue?

Answer: 2 stacks.

Q9. When implementing a Queue using two Stacks (S_1 and S_2), what is the worst-case time complexity for a single deQueue operation if S_2 is empty?

Answer: $O(n)$, because all elements from S_1 must be popped and pushed into S_2 to reverse order.

Q10. What is the maximum number of distinct stack permutations possible with n distinct keys?

Answer: It is given by the n^{th} Catalan Number: $\frac{1}{(n+1)} {}^{(n)}C_n$.

Q11. For $n = 3$ elements, how many unique valid stack pop sequences can be generated?

Answer: $\frac{1}{(3+1)} {}^6C_3 = \frac{20}{4} = 5$.

Q12. True or False: A stack can be used to check whether an expression has balanced parentheses.

Answer: True.

Q13. In a linked list implementation of a stack, where should the push and pop operations be executed to maintain $O(1)$ time complexity?

Answer: At the head (front) of the linked list.

Q14. What is the time complexity of reversing a string using a stack?

Answer: $O(n)$ time and $O(n)$ auxiliary space.

Q15. If a stack function satisfies $f(\emptyset) = 0$ and $f(\text{push}(S, i)) = \max(f(S), 0) + i$, what is $f(S)$ for elements 2, -3, 5 pushed in order?

Answer: 5. Detailed step-by-step verification trace:

- $f(\emptyset) = 0$
- $\text{push}(2) \rightarrow \max(0, 0) + 2 = 2$
- $\text{push}(-3) \rightarrow \max(2, 0) + (-3) = 2 - 3 = -1$
- $\text{push}(5) \rightarrow \max(-1, 0) + 5 = 0 + 5 = 5$

2. Expression Conversion & Evaluation

Q16. Convert the infix expression $a + b * c$ to its equivalent postfix form.

Answer: $a b c * +$

Q17. What is the postfix equivalent of the expression $(A + B) * (C - D)$?

Answer: $A B + C D - *$

Q18. Evaluate the following postfix expression using a stack: $6 \ 3 \ 2 \ + \ * \ 5 \ -$.

Answer: 25. Evaluation flow: $3 + 2 = 5 \rightarrow 6 * 5 = 30 \rightarrow 30 - 5 = 25$.

Q19. While converting an infix expression to a postfix expression using Dijkstra's Shunting-yard algorithm, where are the operators temporarily stored?

Answer: In an internal operator stack.

Q20. What is the prefix form of the infix expression $A * B + C / D$?

Answer: $+ * A B / C D$

Q21. Which parsing style reads an expression from right to left to evaluate it directly with a stack?

Answer: Prefix expression evaluation.

Q22. Evaluate the prefix expression $+ \ * \ 2 \ 3 \ 5$.

Answer: 11. Evaluated right to left: $* 2 3 = 6$, then $+ 6 5 = 11$.

Q23. What is the correct postfix form of $A \wedge B \wedge C$ (assuming right-associativity for $\wedge$)?

Answer: $A B C \wedge \wedge$

Q24. What type of stack underflow/overflow check is needed during postfix expression evaluation when an operator is encountered?

Answer: The stack must contain at least 2 operands to safely pop and evaluate the binary operator.

Q25. Convert the prefix expression $* \ + \ A \ B \ - \ C \ D$ into an infix expression.

Answer: $(A + B) * (C - D)$

Q26. Why are prefix and postfix expressions preferred over infix expressions in computing compiler design?

Answer: They completely eliminate the need for parentheses or multi-tier operator precedence parsing configurations.

Q27. What is the maximum number of operators that can reside in the operator stack during the conversion of $A + B * C - D$?

Answer: 2 operators (+ and * simultaneously occupy the stack before encountering -).

Q28. What is the postfix notation of $A - B / (C * D ^ E)$?

Answer: A B C D E ^ * / -

Q29. While evaluating $8 / 2 / 3 * 5 +$, what is the top element of the stack after processing the + operator?

Answer: 12. Evaluation: $8 / 2 = 4 \rightarrow 4 * 3 = 12$.

Q30. True or False: The operand order remains exactly the same in Infix, Prefix, and Postfix notations.

Answer: True.

3. Queue Conceptual & Circular Queues

Q31. What is the primary operational principle of a Queue data structure?

Answer: FIFO (First In First Out).

Q32. What is the main structural disadvantage of a linear array-based queue implementation?

Answer: False Overflow. Elements cannot be inserted at the back even if space is available at the front due to monotonically increasing rear indices.

Q33. How does a Circular Queue resolve the false overflow issue of linear queues?

Answer: By wrapping indices back to zero using modulo arithmetic: $rear = (rear + 1) \% MAXSIZE$.

Q34. What is the standard formula to check if a circular queue of size N (using the one-empty-slot strategy) is completely full?

Answer: $(rear + 1) \% N == front$.

Q35. What is the formula to calculate the total number of elements currently present in a circular queue?

Answer: $(rear - front + N) \% N$ (assumes standard layout configuration constraints).

Q36. What is the minimum number of queues needed to efficiently implement a standard Stack?

Answer: 2 queues.

Q37. What is a Deque (Double-Ended Queue)?

Answer: A generalized linear sequence structure where element insertions and deletions can occur at both front and rear boundaries.

Q38. In a priority queue, how are elements dequeued?

Answer: Based on their explicit key priorities, rather than sequential insertion temporal ordering.

Q39. What is the time complexity to insert an element into a circular queue implemented via an array?

Answer: $O(1)$.

Q40. Which architectural queue type allows insertions at only one end but deletions at both ends?

Answer: Output-restricted Deque.

4. Advanced Applications & Algorithms

Q41. If a function takes a Queue Q as an argument, pushes all elements into a Stack S, and then enqueues them back into Q, what is the final state of Q?

Answer: The structural sequential order of elements in Q will be completely reversed.

Q42. Which fundamental graph traversal algorithm relies inherently on a Queue data structure?

Answer: Breadth-First Search (BFS).

Q43. Which graph traversal algorithm relies explicitly on a Stack data structure (or system recursion)?

Answer: Depth-First Search (DFS).

Q44. What is the time complexity of the best known algorithm to sort a queue of size n using k external stacks?

Answer: $O(n \log k)$.

Q45. Which data structure is optimal for tracking asynchronously transferred data between two processing entities (e.g., IO buffers)?

Answer: Queue.

Q46. What is the time complexity of checking if a string of length n is a palindrome using one stack and one queue?

Answer: $O(n)$.

Q47. In a monotonic stack pattern, what property does the stack maintain?

Answer: Elements are strictly kept in either increasing or decreasing order to optimize next greater / smaller element linear scans.

Q48. Can a single queue be used to implement a stack smoothly if we allow rotation of elements? If yes, what is the time complexity of the push operation?

Answer: Yes; the push operation takes $O(n)$ time by enqueueing the element and sequentially rotating the previous $n-1$ elements behind it.

Q49. What is the space complexity of a Level-Order tree traversal using a queue on a perfect binary tree with N leaf nodes?

Answer: $O(N)$, which is directly proportional to the maximum layer width of the tree.

Q50. If n inserts and m deletes ($m \leq n$) are performed on an empty queue, how many elements remain?

Answer: $n - m$ elements.

Exam Revision Strategy Note

Ensure you double-check the precise problem-specific initialization of `front` and `rear` pointers when tracking circular space array counts during exam scenarios. Catalan terms scale rapidly: $N(1)=1$, $N(2)=2$, $N(3)=5$, $N(4)=14$, $N(5)=42$.